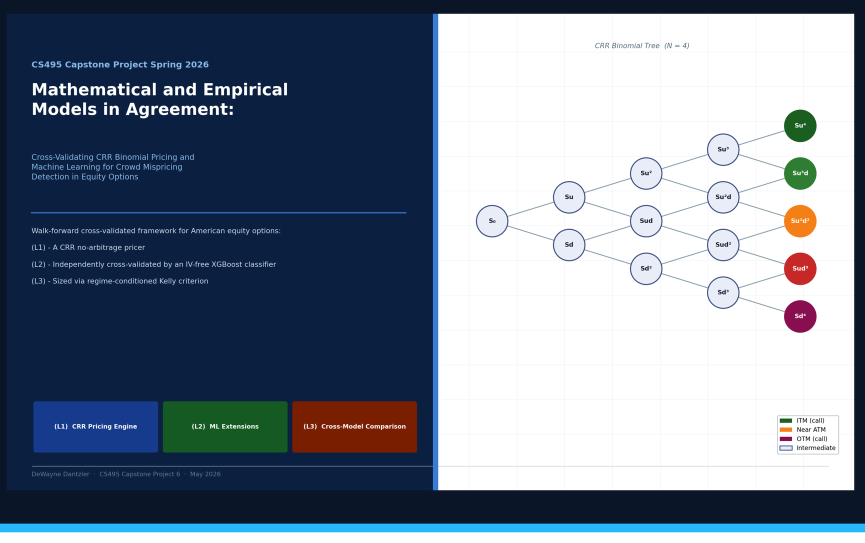
Mathematical and Empirical Models in Agreement

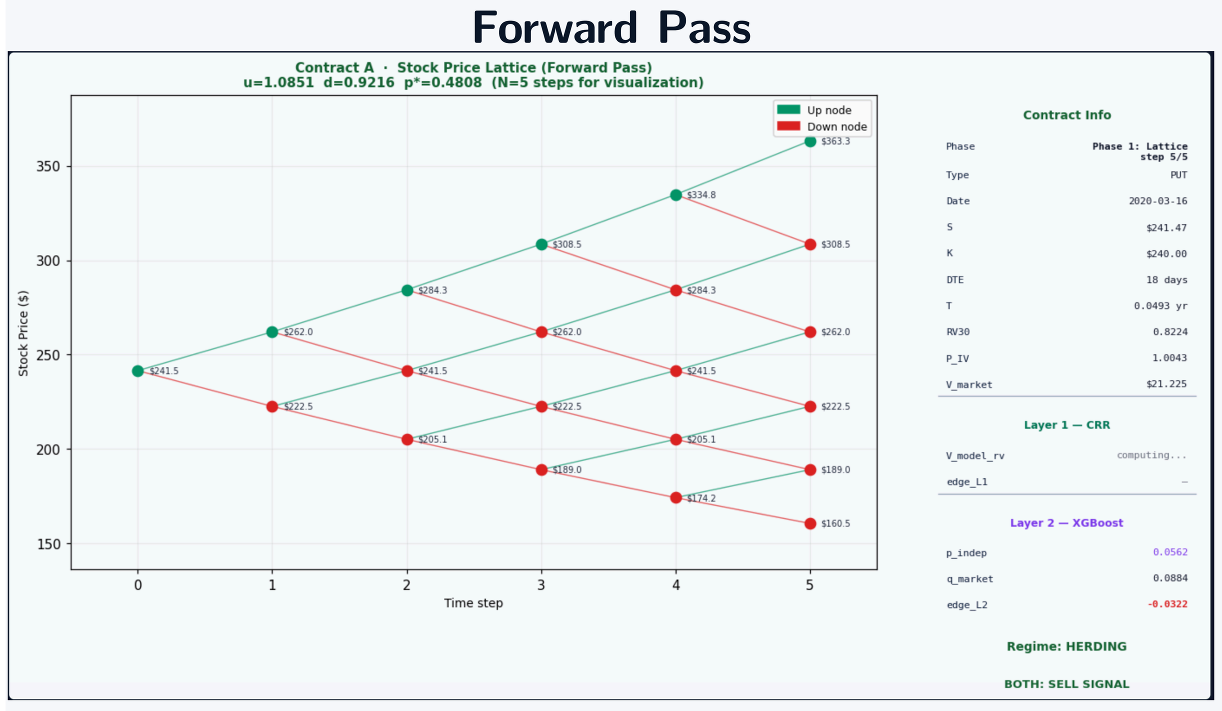
Cross-Validating CRR Binomial Pricing and Machine Learning
for Crowd Mispricing Detection in Equity Options

DeWayne Dantzler · Bellevue College · CS 495 Capstone Project 6 · 2026



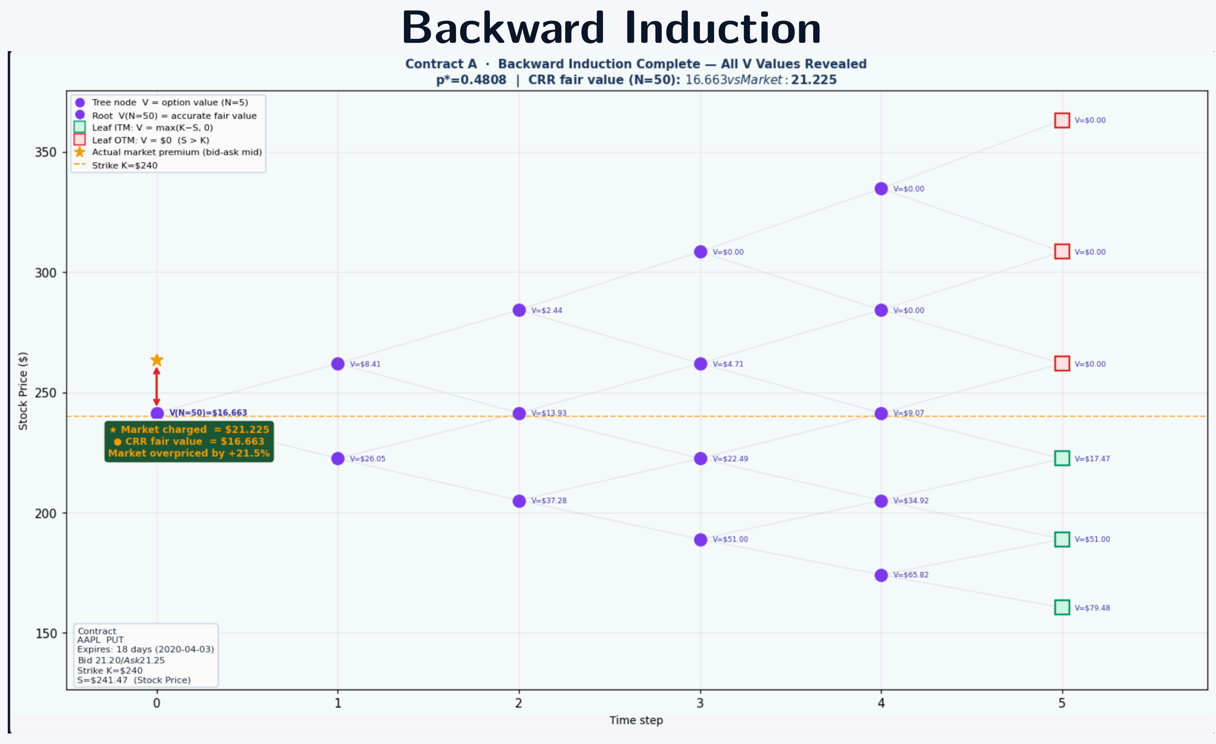
Methodology in Action: Contract A Forward + Backward

Forward Pass



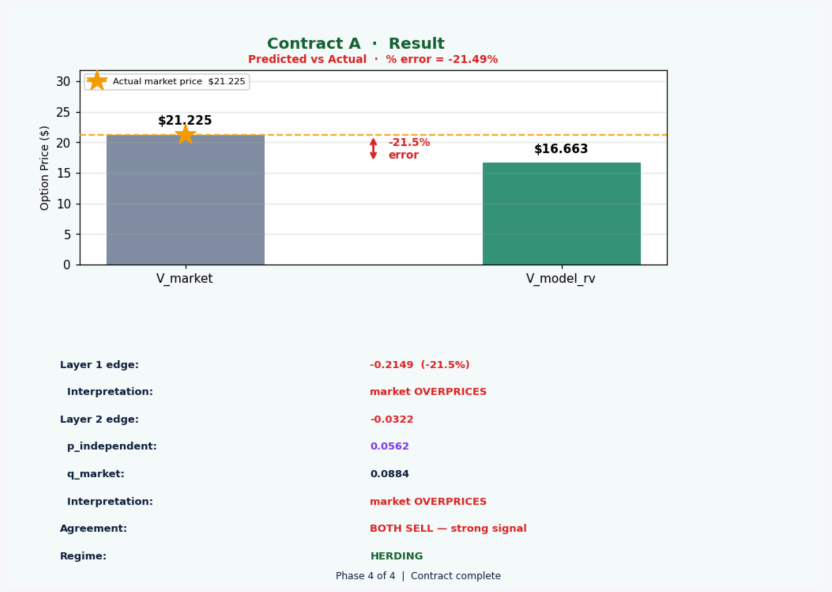
Stock-price lattice using $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p^* = 0.481$

Backward Induction



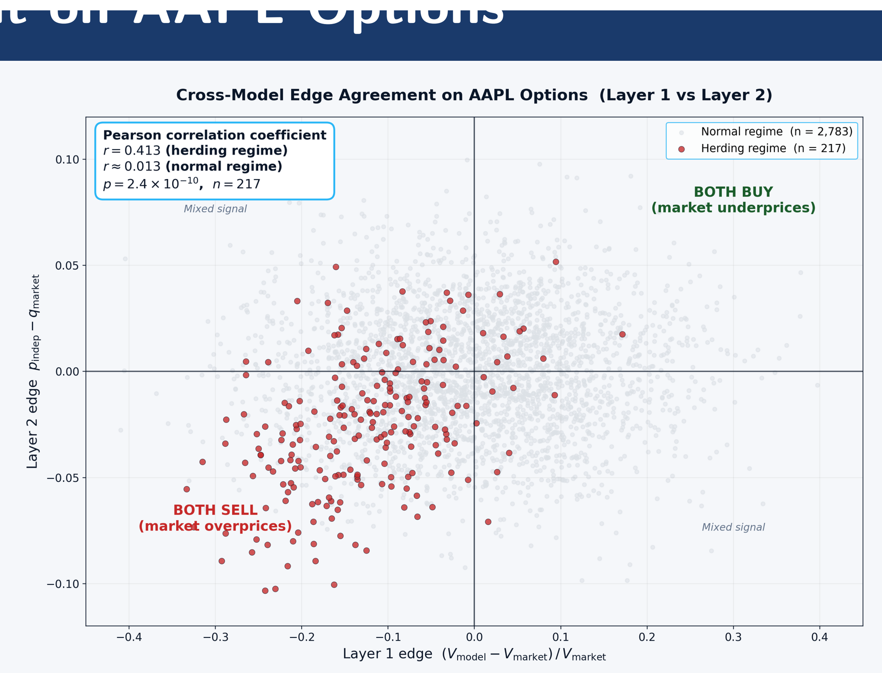
At every node: $V = \max\left(\text{intrinsic},\ e^{-r\Delta t}(p^* V_u + q V_d)\right)$

Contract A Result: Two Independent Models Agree



Layer 1 (CRR) edge $= -21.5\%$ and Layer 2 (XGBoost) edge $= -3.2\%$: **both models say SELL** on this real AAPL put. Same conclusion reached by completely independent methods.

Cross-Model Edge Agreement on AAPL Options



Pearson correlation coefficient $r = 0.413$ (herding) vs $r \approx 0$ (normal), $p = 2.4 \times 10^{-10}$. Agreement concentrated in the herding regime — the empirical fingerprint of Samuelson macro-inefficiency.

Contribution & Methodology (Slide 4)

Contribution & Methodology

Novel Contribution

- **First** to cross-validate CRR against an IV-free ML classifier (fully independent signals)
- Regime-conditioned Kelly sizing from IV/RV_{30} ratio
- Walk-forward 60/20/20 split; zero lookahead bias
- Open-source Streamlit calculator with live chain

Three-Layer Pipeline

L1 CRR Binomial Pricer

↓

L2 XGBoost ITM/OTM Classifier

↓

L3 Kelly Criterion Position Sizer

Data AAPL 2016–2020, 101,535 contracts (walk-forward); AMD April 2026 live chain (live validation).

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Live Demo: Interactive CRR Binomial Pricing Calculator

Project Summary

The question. When an options pricing model produces a theoretical fair value that diverges from the observed market premium, is that divergence large enough — and reliable enough — to justify a trade? If so, how much of the available capital should be committed, given the inherent uncertainty of the model?

Why it matters. Options markets aggregate beliefs into prices the same way prediction markets do, but behavioral forces (herding, vol-risk premium, out-of-the-money skew) distort those prices in predictable ways. The gap between a model's fair value and the observed premium is potentially exploitable by a disciplined, model-driven strategy.

Approach. A Python implementation of the Cox–Ross–Rubinstein (CRR) *American* binomial option pricer is built and validated against live AMD option chain data captured from Fidelity in April 2026. The pricer's output is treated as an independent probability estimate and combined with Kelly Criterion sizing for disciplined risk-controlled trade allocation.

Cross-validation. Two completely independent models are deployed in parallel:

- **Layer 1 (CRR no-arbitrage math)** — theoretical fair value derived from realized volatility (σ = RV30).
- **Layer 2 (XGBoost ML classifier)** — empirical ITM/OTM probability estimate, trained on 101,535 historical AAPL contracts.

Their *agreement* on which contracts the market misprices constitutes mutual validation: neither layer is trained on the other, so concurrent signals come from shared underlying reality.

Key result. Across 3,000 sampled AAPL contracts, the two layers' edge signals agree strongly in the **herding regime** (Pearson $r = 0.413$, $p = 2.4 \times 10^{-10}$) and not at all in the **normal regime** ($r \approx 0$). Agreement is concentrated precisely where behavioral-finance theory predicts crowd bias is strongest — the empirical fingerprint of Samuelson's macro-inefficiency thesis.

Datasets. AMD live chain (April 2026, 4 trade tickets at $\sigma \approx 64\%$) · AAPL 2016–2020 Kaggle (101,535 contracts).

Tech stack. Python 3.13 · NumPy · pandas · XGBoost · scikit-learn · matplotlib · Streamlit · yfinance · Poetry · LaTeX/Beamer · Git / GitHub.

Conclusions.

- Mathematics and ML, working independently, reach the *same* mispricing conclusion in the regime theory predicts.
- The framework is a falsifiable test of EMH (Fama 1970): in a Fama-efficient market, this agreement could not persist.
- Kelly sizing converts the detected edge into capital allocation with controlled drawdown.
- Fully reproducible: one `make run-all` command regenerates every artifact end-to-end.

Key references: Black & Scholes (1973) · Cox–Ross–Rubinstein (1979) · Fama (1970) · Carr & Wu (2009) · Kelly (1956) · MacLean–Thorp–Ziemba (2010)

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Full report and code repository: github.com/dantzlerdc/CS495-Deep-Scholar